3. DESIGN APPROACH

SmartHat is designed to revolutionize personal protective equipment in hazardous industries by integrating advanced sensing and communication technologies into a standard hard hat. SmartHat aims to address the persistent issue of limited hazard awareness and delayed response times in environments such as construction, mining, and manufacturing. By integrating environmental sensors, biometric monitoring, and wireless communication into one wearable device, SmartHat enhances worker safety through early detection and immediate alerting. The project is driven by critical requirements such as accurate detection of carbon monoxide concentrations of 50 ppm or higher within 5 seconds, impact detection of forces exceeding 35 g, continuous 8-hour operation per charge, and ergonomic comfort within a total weight of 3.2 lbs. These technical goals are bound by constraints including a total budget of $1,000 and two-semester-long development timeline. To ensure safety and reliability, SmartHat conforms to ANSI Z89.1 standards for industrial head protection, NEMA 4 enclosure standards for water and dust resistance, and 3GPP Release 13 standards for cellular SMS communication. Together, these requirements, constraints, and standards shape how SmartHat transitions from a concept into a fully functional, safety-enhancing prototype.
The following section provides a comprehensive overview of SmartHat, including its design options, multi-level system diagrams, subsystem architecture, detailed requirements, and overall product summary.

3.1. Design Options

Throughout the design process, multiple configurations were considered for sensor placement, data processing, and communication methods. The final design was selected for its balance between performance, comfort, and manufacturability, ensuring that the hat remains practical for daily use in demanding environments.

3.1.1. Design Option 1

Option 1 focused on developing a fully integrated hard hat with all electronic components and sensors built directly into the shell. This design includes built-in cellular communication, allowing the system to operate completely on its own without requiring any user setup or external device connection. The hard hat automatically collects data, monitors environmental conditions, and sends alerts when unsafe situations are detected. This approach was ultimately chosen because it provides seamless user experience by reducing the chance of user error and ensuring the system always remains active and reliable. By eliminating the need for manual pairing or configuration, this design offers greater convenience, improved safety, and a more professional, ready-to-use product.

3.1.2. Design Option 2

Option 2 considered a removable universal attachment that could be mounted onto a standard hard hat. This design would use Bluetooth connectivity with a smartphone app, allowing the user to view data and receive alerts through their phone. While the design offered flexibility and lower cost, it depended on the user to pair and maintain the connection, increasing the chance of setup errors or missed alerts if Bluetooth was lost or not established. Because a safety device needs to work the moment it is worn, with no extra steps, this dependency reduced reliability, so the option was not selected.

3.2. System Overview

Below are two system overview graphics of SmartHat. Figure 3-1 is a high-level look at the basic inputs and outputs of SmartHat. Figure 3-2 is a more in-depth look, as it shows each subsystem, power inputs, and each output that the system provides to protect the user.

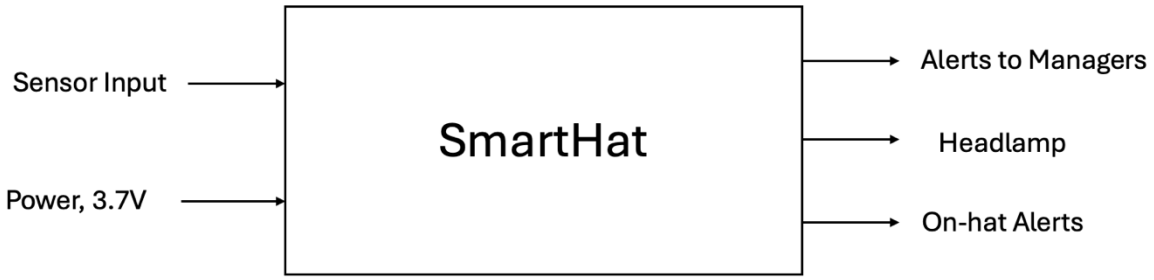

Fig. 3-1: SmartHat System at a Glance (Level 0)

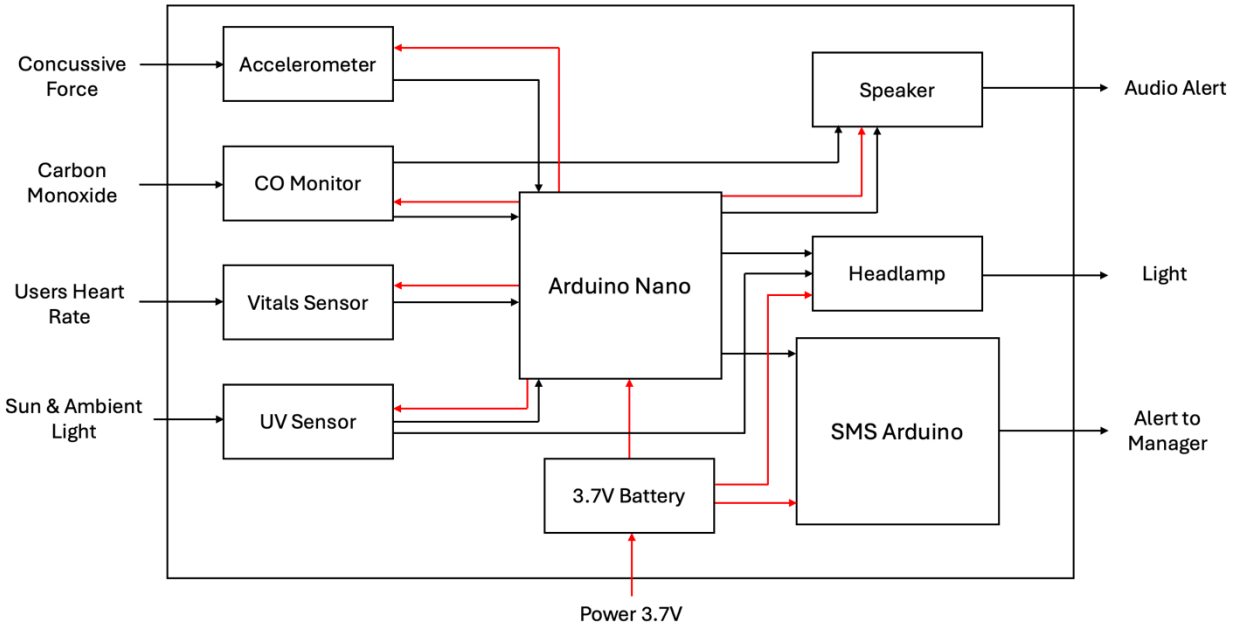

Fig. 3-2: SmartHat Functionality (Level 1)

Figure 3-1 shows the basic overview of inputs and outputs that SmartHat features. This includes information gathered from sensors, power supply, auto-adjusting headlamp, and alerts to the user and managers. Figure 3-2 shows these features more in-depth, going over each subsystem. Power is provided to the sensors and Arduinos by a 3.7V battery. The headlamp will draw its power directly from the battery. The four subsystems on the left will gather data and send it to the Arduino Nano. The Arduino Nano will take those inputs and produce an output to either the user on the hat, or a safety manager through SMS. The UV subsystem will also monitor ambient light and turn on the headlamp if the light is too low. The Arduino Nano will communicate to the SMS Arduino when specific alerts are flagged. The SMS Arduino will then send an alert to the manager so they can respond to accidents or hazards quickly.

3.2.1. Microcontroller

For our project, to effectively manage both sensing and communication tasks we decided to use two different microcontrollers for SmartHat's design. The table 3-1 below shows the specifics of the two choices.

Table 3-1: Microcontroller

Part Number	Type	Manufacturer	Function	Unit Cost
Requirements	Arduino or Raspberry Pi	N/A	supports all sensors or communication	<120
ABX00027 [13]	Arduino Nano 33 IoT	Arduino	Microcontroller for sensor data processing	$23.90
ABX00019 [14]	Arduino MKR NB 1500	Arduino	supports all sensors or communication	$95.20
SC0915	Raspberry Pi Pico W	Raspberry Pi	Microcontroller for sensor data processing	$7.00

Figures 3-3 and 3-4 show the microcontrollers chosen for our design.

Fig 3-3: ARDUINO MKR NB 1500

Fig 3-4: ARDUINO NANO 33 IOT

In Figure 3-3, the Arduino MKR NB 1500 was selected to handle SMS communication and cellular connectivity. This board integrates a u-blox SARA-R410M module, enabling reliable LTE and NB-IoT connections even in low-coverage environments. Its compatibility with Arduino software allows for easy integration with the Nano 33 IoT, enabling efficient data transfer between the two microcontrollers. This separation of tasks enhances system reliability and modularity, ensuring that sensor performance is unaffected by communication latency or cellular network interruptions.

In Figure 3-4, the Arduino Nano 33 IoT serves as the dedicated controller for all sensor subsystems, responsible for collecting, processing, and transmitting environmental data. It was chosen for its compact form factor, low power consumption, and ease of integration with the Arduino MKR NB 1500. Its ARM Cortex-M0+ processor offers sufficient speed and memory to handle multiple sensors simultaneously, meeting our project's real-time monitoring requirements.

3.3. Subsystems

The SmartHat design is composed of five main subsystems that are listed below.

1. Impact Detection: detects concussive force to the hard hat
2. Vital monitoring: tracks heart rate
3. UV detection/headlamp: monitors UV rating/ automatic light when ambient light is low
4. Gas detection: monitors surrounding carbon monoxide levels
5. Wireless Communication: sensors send alerts via wireless messaging when danger is detected

3.3.1. Impact Detection

This subsystem is designed to detect impacts on the helmet and evaluate whether the force of impact is significant enough to warrant alerting a manager of a potential workplace incident. This functionality will be implemented using an accelerometer, which measures the helmet's acceleration during an impact. The recorded acceleration will be compared against a predetermined threshold, above which the impact is classified as potentially concussive. This threshold has been determined to be approximately ±50 G. The selected accelerometer must be compact, cost-effective, compatible with Arduino integration, and capable of accurately measuring acceleration within a range near ±50 G. Table 3-2 below shows each accelerometer considered.

Table 3-2: Accelerometer Options

Product	Measurement Range (G)	Ease of Integration with Arduino	Operating Voltage (V)	Dimensions (mm)	Price ($)
Requirements	±35 – ±60	-	3.33	–	< 40
STMicroelectronics LSM6DSOX 6-axis IMU [1]	±2, ±4, ±8, ±16	High	1.71 – 3.6	2.5 x 3 x 0.83	11.95
SparkFun Triple Axis Accelerometer Breakout - KX134 [2]	±8, ±16, ±32, ±64	Moderate	1.7 – 3.6	2 x 2 x 0.9	35.95
Analog Devices ADXL375 - High G Accelerometer (+-200g) [3]	±200	High	2 – 3.6	25.6 x 17.7 x 4.6	24.95
Analog Devices ADXL1002 - Low Noise, High Frequency MEMS Accelerometerm [4]	±50	Moderate	3 – 5.25	5 × 5 × 1.80	90.74

The selected accelerometer, the KX134, is shown in Figure 3-5. Unlike its competitors, it met all design requirements for optimal integration within SmartHat.

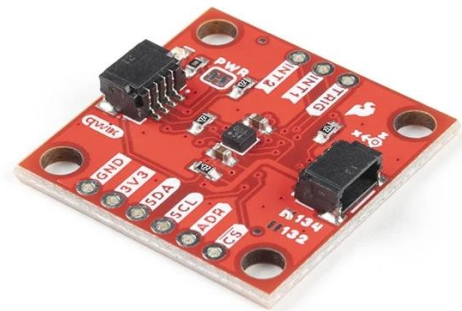

Fig. 3-5: SparkFun Triple Axis Accelerometer Breakout - KX134 [2]

The KX134 is a three-axis accelerometer breakout board capable of measuring acceleration across several user-selectable measurement ranges. This feature provides flexibility during development, allowing the device to be tested and demonstrated under varying acceleration conditions. The KX134 supports both I²C and SPI communication protocols and operates at a nominal voltage of 3.3 V, making integration with Arduino products a straightforward process.

3.3.2. Vitals Monitoring

The vitals subsystem monitors the worker's physical condition, particularly heart rate, to provide early detection of heat stress, overexertion, or other health-related issues in hazardous industrial environments. Real-time data will be transmitted to the system's microcontroller and relayed to safety personnel when abnormal numbers are detected. This is implemented using a heart rate sensor that measures the user's pulse through optical detection. The system continuously compares the recorded heart rate to an already defined threshold and will classify the condition as abnormal if the threshold is exceeded. The threshold that is established is 180 beats per minute. The chosen sensor needs to be small, low-power, compatible with Arduino, and able to measure heart rate accurately within ±5 bpm. The heart rate sensors considered are listed in Table 3-3.

Table 3-3: Vitals Monitor Options

Model	Measurement Method	Accuracy	Supply Voltage	Communication Interface	Price
Requirements	Optical PPG (Red/IR LED)	±5 BPM	3.3 – 5 V	I2C	≤ $30
DFRobot SEN0344	Optical PPG (Red/IR LED)	±3 BPM	3.3 – 5 V	I2C	$15.90
MIKROE-4274 Heart Rate 10 Click	Optical PPG (Red/IR LED)	±3 BPM	3.3 – 5 V	I2C	$48.00
Adafruit Pulse Sensor Amped	Optical PPG (Analog Output)	±5 BPM	3.3 – 5 V	Analog	$25.00

Figure 3-6 shows the DFRobot SEN0344 vitals monitor sensor, which was selected after comparing several options for voltage range, communication method, measurement method, accuracy, and cost.

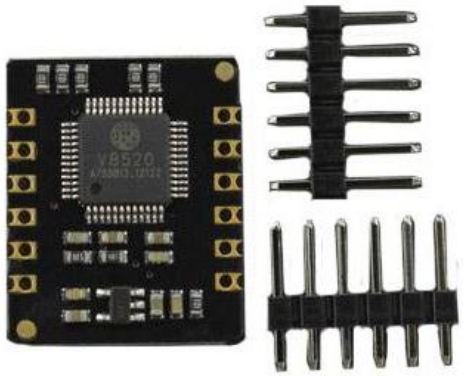

Fig. 3-6: DFRobot SEN0344 Oximeter/Heart Rate Monitor

The SEN0344 stood out from the MAX30102 and Adafruit Pulse Sensor Amped sensors because it offers a flexible 3.3–5 V operating range, operates within the accuracy range, supports I2C communication, provides reliable performance at an affordable price, and consumes minimal amounts of power. It delivers consistent readings of a user's heart rate, making it a strong fit for the SmartHat's vital monitoring features.

3.3.3 UV Detection

The UV detection subsystem is responsible for monitoring both ultraviolet and ambient light levels to ensure user safety and visibility. When ultraviolet radiation reaches harmful levels, the Arduino Nano will output an audible warning through a speaker to alert the user of potential danger. Likewise, when ambient light levels fall below a low lux threshold, the Arduino Nano will automatically turn on the headlamp to improve visibility. This subsystem operates by using a UV light sensor that provides real-time measurements, allowing for quick responses to changing environmental conditions. Multiple sensor options were considered to ensure accuracy, low power consumption, and overall reliability of the subsystem. Table 3-4 lists the sensors evaluated for this subsystem.

Table 3-4: UV Sensor Options

Product	Operating Voltage	Communication Interface	Dimension	Price
Requirement	3.3 VDC	I2C	-	< $15.00
DFRobot SEN0540 [5]	3.3-5 VDC	I2C/UART	1.26x1.06"	$7.90
Adafruit VEML [6]	3-5 VDC	I2C/UART	0.7x0.6"	$5.95
SparkFun AS7331 [7]	2.7-3.6 VDC	I2C	1.00x1.00"	$27.50

Table 3-4 shows the DFRobot SEN0540 UV sensor, which was selected after comparing several options for voltage range, communication method, size, and cost.

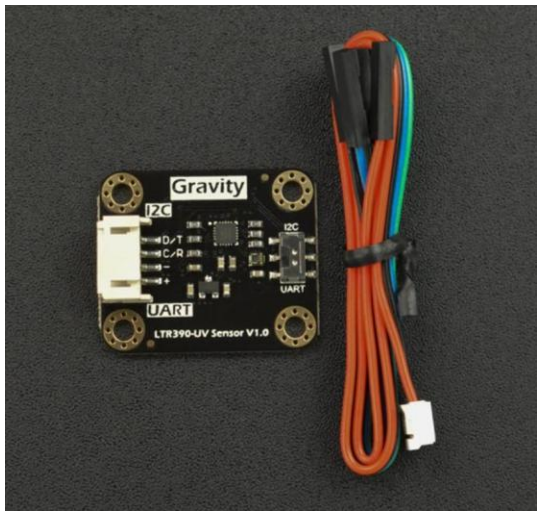

Fig. 3-7: DFRobot SEN0540 [5]

Figure 3-7 shows the ultraviolet sensor selected for the UV Detection Subsystem. The DFRobot SEN0540 offers a simple and convenient design that includes a mounting interface, a wiring harness, and a built-in switch to manually select between I2C and UART communication modes.

Table 3-5 shows the LED headlamp options that were considered for the lighting portion of the subsystem. Each option was evaluated based on voltage rating, brightness, and cost to ensure compatibility with the 5V power supply and adequate illumination for low-light environments.

Table 3-5: LED Headlamp Options

Product	Voltage Rating	Brightness	Price
Requirements	5 VDC	> 200 LM	< $10.00
POCREATION LED Lamp [8]	5 VDC	200 LM	$10.72
PATIKIL LED [9]	5 VDC	200 LM	$8.99
DFRobot LED [10]	5 VDC	300 LM	$3.90

The DFRobot LED, shown in Fig. 3-8, was selected due to its higher brightness of 300 lumens and significantly lower cost compared to the other options, making it the most efficient and practical choice.

Fig. 3-8: DFRobot LED [10]

The DFRobot LED strip operates at 5 VDC and contains 16 COB LEDs with a silicone-coated surface for durability and extended lifespan. It provides bright, uniform illumination and features soldering pads for simple wiring and installation.

3.3.4 Gas Detection

The Gas Detection Subsystem continuously monitors ambient carbon monoxide (CO) levels to protect workers in hazardous environments. Using the DFRobot SEN0466 electrochemical CO sensor, the subsystem accurately detects concentrations ranging from 0 to 1,000 parts per million (ppm) with an accuracy of ±5 ppm, meeting the system requirement of identifying concentrations of ≥50 ppm within 5 seconds. The sensor outputs an analog signal proportional to the CO level, which is processed by SmartHat's Arduino-based control unit. A calibration routine runs at startup to account for baseline environmental CO levels, ensuring accurate readings across varying conditions. When the detected concentration exceeds the OSHA-defined exposure limit of 50 ppm, the system immediately issues multiple alerts. An audible warning is produced through a Visaton 8 Ω speaker, driven by an Adafruit Audio FX Mini Sound Board, both mounted securely within the helmet assembly, along with an SMS notification sent via the communication subsystem to supervisors or safety personnel. This multi-channel warning system ensures that both the user and monitoring staff are promptly informed of potential danger, allowing for swift response and prevention of CO-related incidents. Table 3-6 shows the CO sensor options considered.

Table 3-6: CO Sensor Options

Sensor Model	Detection Method	Range (ppm)	Response Time	Accuracy	Price
Requirements	n/a	10-1000	<10s	±5 ppm	< $100
MQ-7 [15]	Semiconductor	10–1000	60 s	±10%	$6.00
ZE07-CO [16]	Electrochemical	0–1000	≤30 s	±5 ppm	$39.00
SEN0466 [17]	Electrochemical	0–1000	≤5 s	±5 ppm	$66.90

Figure 3-9 shows the SEN0466 CO sensor that was selected after evaluating several detection options based on range, response time, accuracy, and cost. SEN0466 was chosen over MQ series sensor due to faster response time and greater stability under varying humidity and temperature conditions—factors that are critical for reliability in working environments. Its compact, high-precision design allows for accurate readings within the required time and range, while fitting securely onto SmartHat without compromising comfort or balance.

Fig. 3-9: DFRobot SEN0466

The cylindrical component (top left) houses the electrochemical sensing element responsible for detecting carbon monoxide gas, while the circuit board (bottom) manages signal conditioning and analog output. The Grove cable (right) provides easy connection to the Arduino.

Figure 3-10 shows the speaker chosen for the audio alert system. This compact speaker has clear sound, low power consumption, and compatible with the mini sound board.
Figure 3-11 shows the mini sound board that will be paired with the speaker. This sound board stores and plays preloaded alarm tones without requiring a separate amplifier or complex programming, making it a compact and reliable audio alert solution

Fig 3-10: SPEAKER 8OHM 1W [18]

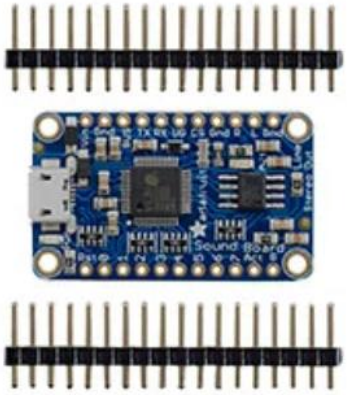

Fig 3-11: FX MINI SOUND BOARD [19]

In conclusion, the Gas Detection Subsystem effectively integrates high-precision sensing, rapid response, and reliable alerting to ensure worker safety in hazardous environments. By combining the DFRobot SEN0466 CO sensor with an Arduino controller, speaker, and FX Mini Sound Board, the system delivers accurate and timely warnings of dangerous CO levels. This design meets all detection and response requirements, providing a dependable safeguard against carbon monoxide exposure in the SmartHat system.

3.3.5 Wireless Communication

The wireless communication subsystem is responsible for enabling remote data transmission from the sensor subsystems to the user's safety managers. Its main function is to provide real-time monitoring and send alerts to safety representatives when an accident occurs or a hazard is present in the workplace.

We needed an additional microcontroller for this subsystem. Specifically, one that had cellular communication capability. Table 3-7 shows the microcontroller options considered.

Table 3-7: Communication Microcontroller Options

Product	Voltage	Communication Interface	Price
Requirements	3.3-5V	UART/I2C/SPI	<$100
SIM900 GPRS/GSM Shield	5V	UART	$27.62
ARDUINO MKR NB 1500	3.3V	UART/I2C/SPI	$95.20
DFRobot Global4G IoT Module	3.3V	UART/AT	$45.00

The Arduino MKR NB 1500, shown in Figure 3-12, was selected after evaluating several options based on capability, reliability, and ease of integration with other subsystems. The MKR NB 1500 has easy integration with our other microcontroller, which is also an Arduino. It has reliable long-range coverage using Narrowband IoT and LTE Cat M1 cellular connectivity. Also, the MKR NB 1500 has low power consumption.

Fig. 3-12: Arduino MKR NB 1500 [13]

The Arduino MKR NB 1500 features a SIM card slot to enable cellular communication, as well as transmit and receive ports to communicate with sensors and other microprocessors.

3.4 Level 2 Prototype Design

The prototype design of SmartHat incorporates an enclosure to house all components that do not require direct exposure to the external environment. The enclosure will achieve a minimum ingress protection rating of IP54, safeguarding the electronic components from dust and moisture-related damage. External connections will be routed through a sealed cable pass-through, utilizing an O-ring to maintain environmental protection. A desiccant, such as a silica gel packet, will be placed inside the enclosure to

prevent internal moisture accumulation. The batteries, Arduino boards, and accelerometer will be contained within this housing.

3.4.1 Level 2 Diagram

Figure 3-13 illustrates the final prototype design of SmartHat. The dotted, colored boxes encircle components belonging to specific subsystems. Black arrows denote signal pathways, while red arrows indicate power distribution. The outermost black rectangle represents the complete SmartHat product, and the inner black rectangle labeled "Enclosure" defines the boundary of the ingress-protected housing. Sensor data is transmitted to the Arduino Nano, which processes the measurements and activates the output components as needed. The Arduinos are provided with a 3.7 V operating voltage and supply 3.3 V to the connected components, with the exception of the LED headlamp, which is powered directly by the battery, but stepped up from 3.7V to 5V via a DC to DC converter.

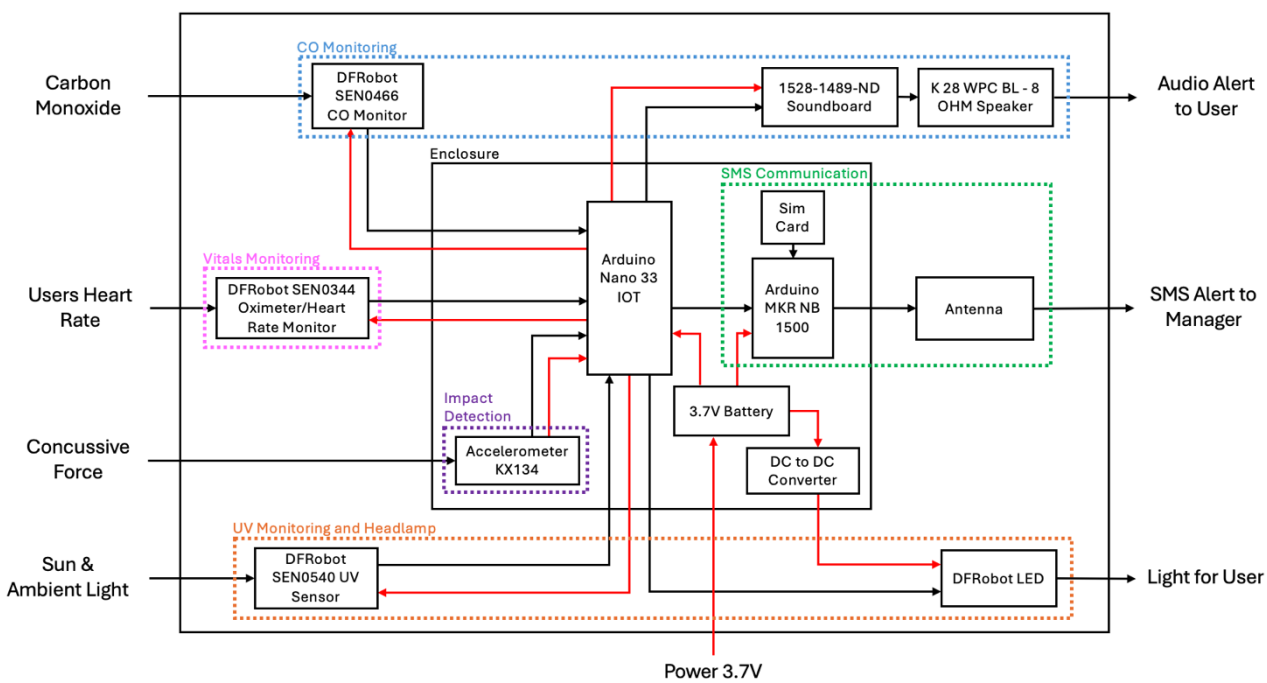

Fig. 3-13: Diagram for SmartHat (Level 2)

The finalized SmartHat design integrates all five subsystems into a unified, dependable safety solution capable of continuous monitoring and prompt alert response. Each subsystem works in tandem with the Arduino Nano 33 IOT to ensure accurate sensing, efficient data handling, and dependable communication while remaining low-power and ergonomically designed. The Level 2 Diagram demonstrates how SmartHat achieves its engineering goals of durability, functionality, and worker protection in hazardous environments. This design approach serves as a solid foundation for future refinement and real-world implementation, achieving the project's goal of improving occupational safety through intelligent, sensor-based innovation.

3.5 References

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